



01/2024:20235 PROCEDURE

2.2.35. OSMOLALITY

PRINCIPLE

GENERAL

Osmolality is a measure of the total number of chemical entities per kilogram of solvent, and thus provides an indication of the osmotic pressure of the solution. Osmolality is dependent on the molal concentration of the solute(s) in the solution, on their dissociation and on the deviation of the solution from ideal behaviour (Raoult's law).

In the International System (SI), osmolality is expressed in moles per kilogram, however the non-SI unit of osmoles per kilogram (osmol/kg) and its submultiple, milliosmoles per kilogram (mosmol/kg), are more commonly used. The unit osmole is defined as the number of moles of solute that contribute to the osmotic pressure of a solution.

The osmolality (ξ_m) of a solution containing i solutes is given by the expression:

$$\xi_m = \sum v_i m_i \phi_{m,i}$$

- v_i = number of entities formed by the dissociation of one molecule of the i^{th} solute; if the solute is non-ionic (non-dissociating), v_i equals 1;
 m_i = molality of the i^{th} solute in the solution, in moles per kilogram of solvent;
 $\phi_{m,i}$ = osmotic coefficient of the i^{th} solute at the molality m_i ; a dimensionless factor.

The osmotic coefficient (ϕ) is a measure of the deviation of the solution from ideal behaviour. For an ideal solution, its value is equal to 1 and osmolality equals molality.

In the case of a real, non-ideal solution, the osmotic coefficient is influenced by the interactions occurring amongst the components (i.e. molecules, ions, solvent) of the solution. The more complex the composition of the solution, the harder it becomes to determine this coefficient.

For this reason, the measurement of a colligative property such as the freezing-point depression is used as a practical means of determining osmolality by obtaining an overall measure of the contribution of the various solutes present in a solution.

PRINCIPLE OF MEASUREMENT

Unless otherwise prescribed in the monograph, osmolality is determined by measuring the freezing-point depression (ΔT_f) of a solution. The relationship between osmolality and freezing-point depression is given by the expression:

$$\Delta T_f = K_f \xi_m$$

where K_f is the molal cryoscopic constant, which is solvent-dependent. For water, the value of K_f is about 1.86 K·kg/osmol (i.e. adding 1 mol of a non-dissociating solute to 1 kg of water results in a decrease in freezing point of about 1.86 K).

EQUIPMENT

An osmometer for freezing-point depression measurement typically consists of:

- an appropriate measuring vessel;
- a means of cooling the sample;
- a temperature-sensitive resistor (thermistor), with an appropriate current or potential difference measurement device that can indicate a temperature depression or give osmolality values directly;
- a means of inducing solidification when supercooling occurs.

CALIBRATION AND ADJUSTMENT

On many instruments, calibration and adjustment are combined in a single procedure that is often referred to as calibration only; however, it is in fact a two-step approach. The first step (calibration) consists of comparing the measured and expected osmolality values of the reference solutions, while in the second step the instrument is adjusted to make the measured and expected values match. The whole procedure is run when necessary (e.g. deviation greater than a defined acceptance criterion) or at appropriate intervals as defined in the quality management system of the user.

Follow the manufacturer's instructions for calibrating and/or adjusting the instrument, using at least two reference solutions with a known osmolality, such as those listed in Table 2.2.35.-1. Commercially available certified solutions with osmolalities equal or similar to those listed may be used. If the low reference point is at zero, use *water R* as the reference solution.

It is recommended that the chosen reference solutions bracket the intended analytical range of sample measurements (i.e. the adjusted range). Where necessary, samples may be measured outside the adjusted range. In such cases, an additional accuracy check (see below) at a value close to the expected osmolality value of the sample must be carried out to verify the accuracy of the instrument in that range.

ACCURACY CHECK

Perform the accuracy check on the day of use. Confirm the accuracy of the instrument using at least one reference solution that is different from the reference solutions used during calibration and adjustment. Select a reference solution with an osmolality close to the centre of the expected osmolality range of the samples. Two reference solutions bracketing the sample within 300 mosmol/kg are also acceptable. Such an approach may also be acceptable if the intended measuring range lies outside the adjusted range.

Measure the solution(s) at least in triplicate and calculate the mean. The mean value must be within 4 mosmol/kg, or 1 per cent, whichever is greater, of the osmolality of the chosen reference solution. If the instrument fails to meet this acceptance criterion, calibrate and adjust it (see above), and then repeat the accuracy check. After a successful accuracy check, the instrument is ready for use.

REFERENCE SOLUTIONS

The quantities of *sodium chloride R* to be added to 1 kg of *water R* to prepare osmolality reference solutions up to 4000 mosmol/kg are given in Table 2.2.35.-1. Other reference solution concentrations can be prepared as needed.

In order to obtain accurate osmolality values, reference solutions must be prepared gravimetrically by weighing not only the solute, as is usual, but also the solvent. The accuracy of the resulting reference solutions depends largely on the weighing accuracy.

Due to the particle size of the sodium chloride, it may not be possible to weigh exactly the necessary amount to the last decimal place as given in Table 2.2.35.-1. In such cases, in order to increase the accuracy of the resulting preparations, the mass of *water R* to be used can be corrected by the ratio between the mass that was actually weighed and the listed mass of *sodium chloride R* for the intended solution given in Table 2.2.35.-1.

Before preparation, dry an appropriate quantity of *sodium chloride R* at 110 °C for 2 h and allow to cool in a desiccator.

Freshly prepared osmolality reference solutions as specified in Table 2.2.35.-1 may be stored at 2-30 °C in airtight containers (preferably glass) to prevent evaporation, and are not to be frozen. These are typically non-sterile, unpreserved solutions and therefore their limited microbial stability needs to be considered. When using commercial solutions, follow the manufacturer's instructions for use and expiry date. Such

solutions are usually provided in single-use ampoules and should be discarded after use. Open ampoules should be used within the duration recommended by the manufacturer.

01/2024:20246



Table 2.2.35.-1. – Preparation of osmolality reference solutions

Osmolality, ξ_m (mosmol/kg)	Mass of sodium chloride R for 1 kg of water R (g)	Osmolality, ξ_m (mosmol/kg)	Mass of sodium chloride R for 1 kg of water R (g)
50	1.530	1600	51.39
100	3.098	1700	54.58
150	4.683	1800	57.76
200	6.278	1900	60.92
250	7.880	2000	64.08
290	9.165	2100	67.23
300	9.487	2200	70.36
350	11.10	2300	73.48
400	12.71	2400	76.59
450	14.33	2500	79.69
500	15.94	2600	82.78
550	17.56	2700	85.85
600	19.18	2800	88.91
650	20.80	2900	91.95
700	22.41	3000	94.98
750	24.03	3100	98.00
800	25.65	3200	101.00
850	27.27	3300	103.99
900	28.89	3400	106.96
950	30.50	3500	109.92
1000	32.12	3600	112.86
1100	35.35	3700	115.79
1200	38.57	3800	118.70
1300	41.79	3900	121.60
1400	45.00	4000	124.48
1500	48.20	-	-

MEASUREMENT

Prepare the instrument for measurement according to the manufacturer's instructions. Before each measurement, ensure that the thermistor probe is dry and clean, in order to avoid any carry-over or dilution effects.

Following the instrument manufacturer's instructions, choose an appropriate measuring vessel, fill it with the necessary volume of the solution to be examined and initiate the measurement. The instrument indicates when the freezing point has been detected.

Perform all measurements using the sample volume that was used during calibration and adjustment. Depending on the type of instrument, the osmolality can be read directly or can be calculated from the measured freezing-point depression.

Preparation of multiple vessels in advance is not recommended because of potential evaporation of the solvent from the sample, resulting in an increase in osmolality values.

2.2.46. CHROMATOGRAPHIC SEPARATION TECHNIQUES⁽¹⁾

INTRODUCTION

Chromatographic separation techniques are multi-stage separation methods in which the components of a sample are distributed between 2 phases, one of which is stationary, while the other is mobile. The stationary phase may be a solid or a liquid supported on a solid or a gel. The stationary phase may be packed in a column, spread as a layer, or distributed as a film, etc. The mobile phase may be gaseous or liquid. The separation may be based on adsorption, mass distribution (partition), ion exchange, etc., or may be based on differences in the physico-chemical properties of the molecules such as size, mass, volume, etc.

This chapter contains definitions and calculations of common parameters and generally applicable requirements for system suitability. Principles of separation, equipment and methods are given in the corresponding general chapters:

- ◇– paper chromatography (2.2.26);
- thin-layer chromatography (2.2.27);
- gas chromatography (2.2.28);
- liquid chromatography (2.2.29);
- size-exclusion chromatography (2.2.30).◇

DEFINITIONS

The system suitability and acceptance criteria in monographs have been set using parameters as defined below. With some equipment, certain parameters such as signal-to-noise ratio and resolution can be calculated using software provided by the manufacturer. It is the responsibility of the user to ensure that the calculation methods used in the software are equivalent to the requirements of the European Pharmacopoeia and to make any necessary corrections if this is not the case.

Chromatogram

A graphical or other representation of detector response, effluent concentration or other quantity used as a measure of effluent concentration, versus time or volume. Ideally, chromatograms are represented as a sequence of Gaussian peaks on a baseline (Figure 2.2.46.-1).

- V_M = hold-up volume;
- t_M = hold-up time;
- V_{R1} = retention volume of peak 1;
- t_{R1} = retention time of peak 1;
- V_{R2} = retention volume of peak 2;
- t_{R2} = retention time of peak 2;
- W_h = peak width at half-height;
- W_i = peak width at the inflexion point;
- h = height of the peak;
- $h/2$ = half-height of the peak.

Distribution constant (K_0)

In size-exclusion chromatography, the elution characteristics of a component in a particular column may be given by the distribution constant (also referred to as distribution coefficient), which is calculated using the following equation:

(1) This chapter has undergone pharmacopoeial harmonisation. See chapter 5.8. Pharmacopoeial harmonisation.